BHANZU GROWTH PM CASE STUDY — DEMO ATTENDANCE

CANDIDATE	ROLE	COMPANY
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Package: this document · working prototype (`dist/index.html` , run instructions in `README.md`) · supporting docs in `docs/`

IF YOU WANT…	READ
The answer to Parts 1–4	This document
What the prototype does and how it was built	`README.md` , `docs/PRODUCT_SPEC.md`
How the test would run and be decided	`docs/EXPERIMENT_PLAN.md`
Event definitions and privacy	`docs/ANALYTICS_PLAN.md`
What was checked, how, and the results	`docs/VERIFICATION_REPORT.md`
What changed from the first version and why	`docs/QUALITY_AUDIT.md`
Short answers to likely challenges	`docs/INTERVIEW_DEFENSE.md`

EXECUTIVE SUMMARY

- **Data.** Attendance is 66% for same-day and next-day bookings and 55% at a 7-day gap, an 11-point spread. The pattern is not a smooth slope: a 5-point step between day 3 and day 4, a plateau at 57%, a rebound to 63% at day 6, and the low at day 7.
- **What it can't tell us.** The table has no booking counts, so we cannot compute confidence intervals, test significance, or reconstruct the overall rate. It is observational, so it cannot show that waiting *causes* no-shows; who books later may differ from who books same-day.
- **The larger problem.** Even the best rows lose one booking in three. The gap explains part of the loss; most of it exists at every gap.
- **Diagnosis.** A decision made in an ad-click moment has to survive forgetting, competing priorities, a child who wasn't part of the decision, doubts about a sales pitch, and joining friction. Today nothing gives the family a reason to re-engage during the wait.
- **Recommendation, in two parts.**

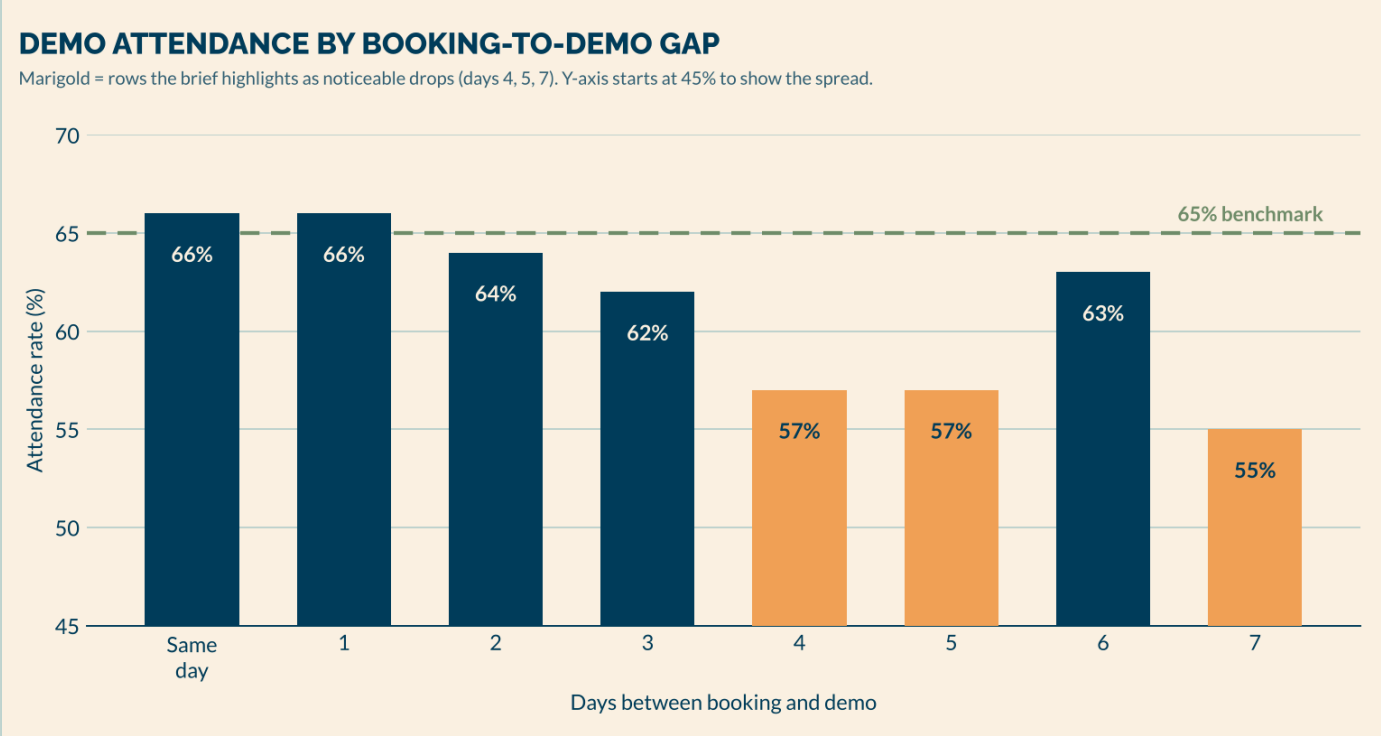
1. **Baseline (ship to everyone, no test needed to justify it):** rewritten confirmation and reminder messages, one-tap rescheduling, calendar and joining help.
2. **Primary learning experiment:** the **Math Mission**, a 5-minute, child-facing warm-up sent with the confirmation, plus a parent preparation hub. It is the most uncertain and potentially most valuable idea, and the only one that works through the child.

- **Measurement.** Primary metric: demo attendance, intent-to-treat, using Bhanzu's existing attendance definition. Leading indicator: **24-hour Math Mission activation rate** among treatment bookings with at least 24 hours to the demo, first read on day 3 of the test.

PART 1 — READING THE DATA

1.1 The data

Every figure below was checked against the brief (table on page 2; rows 4, 5 and 7 are shaded orange in the original). No discrepancies.



DAYS BETWEEN BOOKING AND DEMO	ATTENDED	NO-SHOW	VS 65% BENCHMARK	VS SAME DAY
0 (same day)	66%	34%	+1 pt	—
1	66%	34%	+1 pt	0
2	64%	36%	−1 pt	−2 pts
3	62%	38%	−3 pts	−4 pts
4	**57%**	**43%**	**−8 pts**	**−9 pts**
5	**57%**	**43%**	**−8 pts**	**−9 pts**
6	63%	37%	−2 pts	−3 pts
7	**55%**	**45%**	**−10 pts**	**−11 pts**

Bold rows are the ones the brief shades orange ("attendance dropped noticeably").

1.2 What the data tells us — and what it does not

Observed facts (read directly from the table)

1. Days 0 and 1 are the best rows at 66%, 1 point above the 65% benchmark.
2. Days 2 and 3 fall 2 points each (64%, 62%).
3. Day 3 → day 4 is the largest single step: −5 points.
4. Days 4 and 5 are equal at 57%.
5. Day 6 is 63%, 6 points above days 4–5.
6. Day 7 is the lowest at 55%, 11 points below same-day.
7. No-shows rise from 34% to 45% between day 0 and day 7, about a third more (11 ÷ 34 ≈ 32% relative).
8. The simple average of the eight rows is 61.25%. That is not comparable to the 65% benchmark (see 1.3).

Interpretations (what the facts suggest, with the reasoning shown)

- **The pattern is a step, not a slope.** Days 0–3 lose 4 points across three days; days 3–4 lose 5 in one. Something changes around the 4-day mark. It may be a feature of the calendar (e.g. a weekend falling inside the wait), of the reminder cadence, or of who books four or more days out.
- **The day-6 rebound rules out a simple "people forget over time" story.** Forgetting alone would produce a monotonic decline.
- **The floor matters at least as much as the slope.** The best row loses a third of bookings. A fix aimed only at long gaps leaves most of the loss untouched.
- **The gap is operationally expensive even before we know why.** If booking costs are equal, the cost per *attended* demo is cost per booking ÷ attendance rate, so a 7-day booking costs 66 ÷ 55 = 1.2×, i.e. about 20% more per attended demo than a same-day booking. Per 1,000 bookings, the 7-day rate means about 110 more empty seats than the same-day rate. Those are teacher slots booked and unused, and counsellor capacity planned for families who never arrive.

Hypotheses (plausible explanations, none established)

PATTERN	CANDIDATE EXPLANATIONS
Step at day 4	A weekend falls between booking and demo more often at 4+ days; the current reminder schedule leaves a long silent stretch; nearer slots were full, pushing some families (possibly lower-intent) to day 4+
Day-6 rebound	Weekday alignment: if demos cluster on weekends, 6-day bookings may be disproportionately weekend-to-weekend plans (book Sunday for Saturday) made by deliberate planners; a specific campaign or market; slot supply; noise from a small row
Day-7 low	"Same time next week" bookings that are a soft yes; possibly a default option in the booking UI
34% floor	Impulse bookings, wrong contact details, the child not on board, joining problems, no concrete plan for the hour

Unknowns that block stronger conclusions

- **Booking counts per row.** Without them we cannot compute confidence intervals or test whether rows differ. For scale only: *if* a row had about 200 bookings, a 95% interval would be about ±7 points, enough to make the day-6 rebound indistinguishable from noise. We do not know the counts, so this is an illustration, not a finding.
- **How the 65% is computed.** It is described as "approximately 65%" across all bookings. It may cover a different period or include gaps beyond 7 days.
- **How "attended" is defined** (any join, or a minimum duration) and how cancellations, reschedules and duplicates are counted.

1.3 Unweighted average vs weighted overall rate

The overall attendance rate is a **booking-weighted** average: each row counts in proportion to its number of bookings. The 61.25% simple average gives every row equal weight, so it is not an estimate of the overall rate.

Taken at face value, an overall rate of 65% with every row from day 2 onward at 64% or below would mean **at least half of all bookings sit in the day 0–1 rows** (the 66% rows must pull the average up to 65%). But the figures are rounded to whole percentages. If the true values were, say, 65.5% for days 0–1, 64.49% for day 2 and 64.5% overall, the bound collapses to almost nothing. So the table **suggests**, but cannot show, that short-gap bookings dominate volume. The row counts settle it, and the answer changes prioritization: if long-gap bookings are a small share of volume, gap-specific fixes have a small ceiling.

1.4 Treating 65% as a planning baseline

The brief calls 65% "our current benchmark". I use it only for planning (e.g. illustrative sample-size inputs). The experiment's control rate will be measured in the control arm itself, after the baseline message changes ship, and may differ.

1.5 Breakdowns needed to separate cause from cohort mix

BREAKDOWN	WHAT IT WOULD REVEAL
Gap × **demo weekday** and × **booking weekday**	Whether the day-4 step and day-6 rebound are calendar effects
Gap × **time of day** of the demo	Evening congestion vs weekend slots
Gap × **slot availability** (earliest slot offered vs slot chosen)	Whether later bookers chose to wait or had to (selection vs supply)
Gap × **lead source / campaign / market**	Whether long gaps come from particular audiences or time zones
Gap × **reminder exposure** (delivered, read)	Whether long-gap families simply received fewer or worse-timed messages
Gap × **reschedule / cancel history**	Whether "no-shows" include failed attempts to reschedule
Gap × **join-link clicks vs actual joins**	Technical friction vs lost intent
Gap × **child age band**, first-time vs returning visitor	Segment differences
Attendance → enrollment by gap	Whether a long-gap attendee is worth more or less, which sizes the prize

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PART 2 — DIAGNOSING THE DROP

2.1 Do I agree with the team?

"Attendance falls as the booking gap grows." Partly. Days 0–1 average 66%; days 4–7 average 58%. But the decline is not monotonic, and the gap is likely a proxy for other things: the weekday the demo falls on, how the slot was chosen, and who the family is. I would use the gap to *target* interventions, not treat it as the cause.

"A 5-day booker is no different from a same-day booker." I disagree. Clicking and booking are cheap; attending costs an evening. The chosen date is the family's first signal of how the demo fits their life:

- **Same-day bookers** act on live intent, probably with the child nearby. Intent is high but fragile.
- **Five-day bookers** are a mix: planners who picked a slot that works (possibly *higher* commitment), soft yeses who picked a later date because nothing felt urgent, and families who had no nearer slot.

- **Their context differs**: more days for exams, tuition, illness, travel, or a family member questioning the decision.

So part of the day-5 shortfall is caused by waiting and part by who chooses to wait. A randomized experiment on the waiting period isolates the first part; the breakdowns in 1.5 estimate the second.

2.2 What happens between booking and session

The booking is made in a warm moment ("I want my child to enjoy maths"). Attendance happens in a cold one ("it's 5:30, he's tired, I'm cooking, is this a sales pitch?"). Forces against attendance as time passes:

1. **Memory decay.** One booking among dozens of things that day; without well-timed cues it is forgotten.
2. **Present bias and competing priorities.** On the day, homework, tuition and chores feel more urgent than an optional free class.
3. **Fading emotional intent.** The ad created a feeling; by day 5 only the cost (an hour) is left.
4. **Zero sunk cost.** Nothing has been invested, so skipping costs nothing.
5. **Second thoughts.** Fear of sales pressure, doubts about online maths, an unconvinced family member; more days means more time for doubt and comparison.
6. **The child isn't on board.** The parent booked it; the child may not know, or may resist "extra maths". The child is the one who has to attend.
7. **Logistics.** Which device, which link, will it work. Friction that appears in the last 15 minutes is often fatal.
8. **Silence.** Nothing between confirmation and a last-minute reminder renews the commitment.

2.3 Hypothesis tree

```
Why do booked families not attend?
├── A. They forget or deprioritize it
│   ├── A1 Forgetfulness
│   └── A2 Competing priorities on the day
├── B. The timing stops working
│   └── B1 Scheduling conflicts that arise after booking
├── C. Motivation fades
│   ├── C1 Loss of urgency
│   ├── C2 No interaction after booking
│   └── C3 The child is not motivated or involved
├── D. Doubt grows
│   ├── D1 Uncertainty about what the demo is
│   └── D2 Low trust / fear of a sales pitch
├── E. They try but cannot join
│   └── E1 Technical and joining difficulties
└── F. They were never very committed
    └── F1 Intent differs between gap cohorts (selection)
```

2.4 Hypotheses, evidence and tests

#	HYPOTHESIS	WHY PLAUSIBLE	EVIDENCE THAT WOULD SUPPORT IT	DATA TO TEST IT	INTERVENTION IT SUGGESTS	IF IT'S WRONG
A1	Forgetfulness	Booking happens mid-scroll; nothing anchors it	No-shows with unread reminders; "I forgot" replies	Reminder delivery/read × outcome; one-question no-show survey	Better-timed reminders; calendar add	We over-message engaged parents; opt-outs rise

#	HYPOTHESIS	WHY PLAUSIBLE	EVIDENCE THAT WOULD SUPPORT IT	DATA TO TEST IT	INTERVENTION IT SUGGESTS	IF IT'S WRONG
A2	Competing priorities	Crowded evenings; a free class is easiest to drop	Lower attendance for weekday evenings and exam weeks	Attendance × slot time, weekday, school calendar	Better default slots; easy rescheduling; plan the hour	Slot changes ops can't staff
B1	Scheduling conflicts	Longer gaps leave more room for new commitments	Reschedule requests rise with gap; "something came up" replies	Reschedule/cancel × gap	One-tap reschedule, offered proactively	Rescheduling becomes indefinite deferral
C1	Loss of urgency	Ad-driven intent decays; fits the day-4 step	Engaged families attend more *at the same gap*	Engagement × attendance, controlling for gap	Engagement during the wait	Engagement may only mark intent (selection)
C2	No interaction after booking	The flow is silent between confirmation and reminder	Families who click/reply to confirmation attend more	Confirmation click/reply × outcome	Something worth opening during the wait	We build content nobody opens
C3	Child not involved	The child attends but rarely chose to	No-show reasons mentioning the child	Survey; attendance × age	Child-facing warm-up with a reason to show up	Children engage but parents still skip
D1	Uncertainty about the demo	Parents don't know what the hour involves	Pre-demo support questions; lower attendance among first-time buyers	Support logs; landing analytics	"What to expect" content	Extra reading with little effect
D2	Fear of a sales pitch	Free demo + counsellor call is a familiar pattern	No-shows concentrated in low-familiarity segments	Survey; source segmentation	Honest framing of the counsellor call	Softer framing may lower later conversion
E1	Joining difficulties	Lost links, app installs, audio	Link clicks without a join; support tickets in the first 10 minutes	Join funnel: click → join → 10 min	Readiness checklist; T−30 and T+5 messages	Fixes only a small slice
F1	Intent differs by cohort	Later bookers may be lower intent or forced late	Chosen-later and forced-later cohorts attend differently	Slot offered vs chosen; gap × weekday	Show soonest slots first; gap-tailored touchpoints	If intent explains everything, fix top-of-funnel quality instead

2.5 Problem statement

> Parents book a free demo in a moment of emotional intent, then wait up to a week with no reason to re-engage. Forgetting, competing priorities, an uninvolved child, and doubts about the demo erode the decision. About one in three booked demos is lost, and about 45% of demos booked a week ahead.

2.6 Testable hypothesis

> **If** booked families receive a short, child-facing Math Mission with their confirmation, plus a parent preparation hub, **then** demo attendance will be higher than with the improved confirmation and reminders alone, **because** the child gains anticipation and a reason to show up (a badge for the teacher), the parent gets a reason to reopen the booking during the wait, and joining friction is handled before the day. We expect a larger effect at gaps of 3+ days.

PART 3 — IDEAS TO IMPROVE ATTENDANCE

3.1 The options

#	IDEA	WHAT IT IS	BEHAVIOURAL LEVER
1	Rebuilt confirmation and reminder sequence	Personal WhatsApp messages at booking, day before, 30 min before, and T+5 if not joined	Timely cues; implementation intentions; less friction
2	**Math Mission**	5-minute puzzle set by age band; a badge to show the teacher; a summary for teacher and counsellor	Anticipation; small investment (IKEA effect); child as advocate; endowed progress
3	One-tap rescheduling	"Pick a new time" in every message and on the hub	Turns silent no-shows into reschedules; keeps intent alive
4	Parent reassurance and preparation	What happens before, during and after; honest note on the counsellor call	Uncertainty reduction; trust; autonomy
5	Calendar and joining readiness	Add to calendar, device checklist, how to join	Externalized memory; friction removal
6	Gap-aware booking	Show soonest slots first; extra touchpoint for 3+ day gaps	Defaults; shorter decay window
7	Re-engage silent families	A personal check-in if no click/reply within 48h and gap ≥3	Personal accountability; reciprocity
8	Commitment step	"Tap I'll be there" the day before	Commitment and consistency

3.2 Two decisions, two lenses

The first version of this case scored all eight ideas on one 25-point scale and then picked the 18-point Math Mission over the 21-point reminder sequence. That looked contradictory, because the table mixed two separate decisions:

1. **What should we ship now?** The question is "is it safe, cheap and very likely to help?" Confidence and effort dominate.
2. **What should we test first?** The question is "where is the most valuable uncertainty we can resolve quickly?" An idea we are already confident in has little to learn from a test.

The scores below keep the original judgements where they still hold and add the second lens. Every score is a judgement from the brief and general product experience, not measured data; the evidence column says what each rests on.

LENS A — READINESS TO SHIP (DECIDES THE BASELINE)

Criteria, 1–5 each, 5 is best: **Impact** (likely effect on attendance), **Confidence** (how well-established the mechanism is), **Effort** (5 = low), **Risk** (5 = low).

#	IDEA	IMPACT	CONFIDENCE	EFFORT	RISK	TOTAL /20	EVIDENCE FOR THE SCORES	DECISION
1	Reminder sequence	3	4	5	4	**16**	Reminders are standard practice for appointments; copy-only change	**Baseline**
5	Calendar + joining	2	4	5	5	**16**	Removes known friction; link and checklist are trivial to build	**Baseline**
3	One-tap reschedule	3	4	4	4	**15**	Turns some no-shows into reschedules; depends on slot supply	**Baseline**
8	Commitment step	2	3	5	3	**13**	Cheap; can feel transactional	**Baseline** (button inside the day-before message)

#	IDEA	IMPACT	CONFIDENCE	EFFORT	RISK	TOTAL /20	EVIDENCE FOR THE SCORES	DECISION
4	Parent reassurance	2	3	5	5	**15**	Content-only; uncertain size of effect	Folded into the Math Mission hub
2	Math Mission	4	2	3	4	**13**	Only child-facing lever, but no internal evidence families will open it	Test (Lens B)
6	Gap-aware booking	3	2	3	3	**11**	Day-6 pattern unexplained; touches booking conversion	Analyse first
7	Re-engage silent families	3	3	2	3	**11**	Human time per booking; costly at scale	Later, on the silent segment

The Math Mission ranks low on readiness to ship, and that is correct: we should not roll it out to everyone on faith.

LENS B — VALUE OF TESTING FIRST (DECIDES THE EXPERIMENT)

Only ideas *not* already in the baseline compete here (the baseline ships to both arms, so testing it inside this experiment is impossible).

Criteria, 1–5: **Incremental impact if it works** (on top of the baseline), **Learning value** (how much uncertainty the test resolves and how much it would change future decisions), **Signal speed** (how soon a leading indicator appears), **Test cost** (5 = cheap), **Risk** (5 = low).

IDEA	INCR. IMPACT	LEARNING VALUE	SIGNAL SPEED	TEST COST	RISK	TOTAL /25	EVIDENCE FOR THE SCORES
Math Mission (+ hub)	4	5	4	3	4	**20**	Tests whether the child is a lever at all, a question no other idea answers; activation is visible within 24h; prototype already built
Gap-aware booking	3	4	2	2	2	**13**	Needs attendance data to read (no fast proxy); changes the booking page, which risks booking conversion
Re-engage silent families	3	3	3	2	3	**14**	Effect plausible but cost scales with bookings; better tested after the mission identifies who stays silent

Why the Math Mission is the right first test

1. It is the only idea that acts on the child, who is the person who has to attend.
2. It fills the silent window with a reason to come back.
3. It produces a leading signal within a day.
4. It carries into later funnel stages: the teacher can open with the badge; the counsellor can start from what the child did.
5. Content and copy can change without engineering, so iteration is cheap.

Risks and dependencies: families must open it (delivery and copy matter); teachers and counsellors must use the summary for the carry-over benefit; children's data rules constrain what we track (`docs/ANALYTICS_PLAN.md` §6).

When to revisit this decision

- Row counts show long-gap bookings are a large share of volume and slot supply drives the gap → move gap-aware booking up.
- Join-funnel data shows many no-shows clicked the link but failed to join → prioritize joining support.
- WhatsApp delivery or read rates are poor → fix the channel before testing anything delivered through it.

- The mission's 24-hour activation stays very low after two rounds of copy changes → stop and move to re-engagement.

3.3 Sequence

WHEN	WHAT
Week 0	Pull the breakdowns in 1.5; confirm the attendance definition
Week 1	Ship the baseline (messages 1, 3, 4, T+5 rescue, 5; rescheduling; calendar/joining) to all bookings. If leadership wants its effect measured, run it as its own short A/B against today's flow first; do not stack it into the mission test
Week 2+	Launch the Math Mission experiment on top of the baseline

3.4 Message copy

All messages are proposed **WhatsApp Business template** messages. **None is implemented**; the prototype only demonstrates the experience the links open. Production needs template approval, opt-in, quiet hours (no sends 9 pm–8 am local), a per-booking message cap, and a STOP option. `{{fields}}` come from the booking system.

Arm assignment: baseline messages go to both arms. Message 2 and the mission link/button in Message 1 are treatment-only.

MESSAGE 1 — BOOKING CONFIRMATION

- **Timing:** within 1 minute of booking · **Audience:** every new booking
- **Purpose:** confirm details, set the plan, give the child something to do now
- **Mechanism:** uncertainty reduction; immediate reward; plants the mission while intent is warm
- **Fields:** `parent_first_name`, `child_first_name`, `demo_day`, `demo_date`, `demo_time`, `timezone`, `age_band`, `mission_link`, `details_link`
- **Desired action:** tap *Start Math Mission* (treatment) or *Add to calendar*

> Hi {{parent_first_name}} 👋
>
> {{child_first_name}}'s free Bhanzu maths class is booked!
>
> 📅 {{demo_day}}, {{demo_date}}
> ⏰ {{demo_time}} ({{timezone}})
> 💻 60 minutes, live with a teacher
>
> While you wait, {{child_first_name}} can try a 5-minute Math Mission made for ages {{age_band}}. It's a fun warm-up with a secret speed trick, not a test.
>
> 👉 {{mission_link}}
>
> Need a different time? Just reply CHANGE.
>
> [Start Math Mission] [Add to calendar] [Change time]

Control version: identical, without the Math Mission paragraph and button; the link goes to a plain details page.

MESSAGE 2 — EARLY ENGAGEMENT INVITATION (TREATMENT ONLY)

- **Timing:** 24 hours after booking (after the leading-indicator window closes) · **Audience:** treatment bookings with no `mission_started`, demo at least 24 hours away
- **Purpose:** a second reason to open the booking during the wait
- **Mechanism:** curiosity; the child as advocate; small ask
- **Fields:** `parent_first_name`, `child_first_name`, `demo_day`, `mission_link`
- **Desired action:** child starts the mission

> Hi {{parent_first_name}}, a quick one for {{child_first_name}} 🙂
>
> We've made a 5-puzzle Math Mission to warm up before {{demo_day}}'s class. At the end, {{child_first_name}} gets a badge to show the teacher.
>
> It takes about 5 minutes. A good one for after homework 👇
> {{mission_link}}

Variant for families who completed the mission:

> 🏅 {{child_first_name}} earned the {{badge_name}} badge!
>
> Bring it to class on {{demo_day}} at {{demo_time}}; {{child_first_name}} can tell Ms. {{teacher_first_name}} about it at the start.
>
> A 2-minute read on getting the most out of the class: {{prep_link}}

MESSAGE 3 — DAY-BEFORE REMINDER

- **Timing:** 6 pm local the day before (same-day bookings: merged into Message 4) · **Audience:** all confirmed bookings
- **Purpose:** renew commitment, plan the hour, surface conflicts early enough to reschedule
- **Mechanism:** implementation intention (where, which device); rescheduling easier than skipping; one-tap commitment
- **Fields:** `child_first_name`, `demo_time`, `join_link`
- **Desired action:** *I'll be there* or *Change time*

> Tomorrow's the day! 🎉
>
> {{child_first_name}}'s live maths class is at {{demo_time}}.
>
> Three small things help:
> 1️⃣ A laptop or tablet works best (a phone is fine too)
> 2️⃣ A quiet corner, a pencil and paper
> 3️⃣ Join 5 minutes early. The link: {{join_link}}
>
> Something came up? No problem, just pick a new time.
>
> [I'll be there 👍] [Change time]

Hinglish variant

> Kal class hai! 🎉
>
> {{child_first_name}} ki live maths class kal {{demo_time}} baje hai.
>
> Bas teen cheezein:
> 1️⃣ Laptop ya tablet best rahega (phone bhi chalega)
> 2️⃣ Ek shaant jagah, pencil aur copy
> 3️⃣ 5 minute pehle join karein: {{join_link}}
>
> Kuch aa gaya? Koi baat nahi, naya time chun lijiye.
>
> [Hum aayenge 👍] [Time badlein]

MESSAGE 4 — SHORT PRE-DEMO REMINDER

- **Timing:** 30 minutes before · **Audience:** all confirmed bookings
- **Purpose:** the link at the moment of action; catch technical problems early
- **Mechanism:** cue at the point of action; human safety net
- **Fields:** `child_first_name` , `teacher_first_name` , `join_link`
- **Desired action:** tap the link; reply HELP if stuck

> {{child_first_name}}'s class starts in 30 minutes ⏰
>
> Ms. {{teacher_first_name}} is getting ready. Join here: {{join_link}}
>
> Trouble joining? Reply HELP and our support team will get in touch.

T+5 minutes, not joined (live rescue):

> We've started! {{child_first_name}}'s seat is still open. Join here: {{join_link}}
>
> If today doesn't work anymore, reply CHANGE and we'll find another time.

MESSAGE 5 — RESCHEDULING AND RECOVERY

- **Timing:** 30–60 minutes after the class ends, no join recorded · **Audience:** no-shows and anyone who replied CHANGE
- **Purpose:** turn a lost booking into a rebooked one without blame
- **Mechanism:** no guilt; very low effort; continuity; soonest slots first
- **Fields:** `parent_first_name` , `child_first_name` , `slot_1` – `slot_3` , `badge_name` (if earned)
- **Desired action:** reply 1, 2 or 3

> Hi {{parent_first_name}}, we missed {{child_first_name}} in class today. Completely understandable, days with children get busy.
>
> Would one of these times work instead?
>
> 1️⃣ {{slot_1}}
> 2️⃣ {{slot_2}}
> 3️⃣ {{slot_3}}
>
> Reply 1, 2 or 3 and it's booked. {{child_first_name}}'s {{badge_name}} badge is saved for the new class.
>
> If now isn't the right time for classes, reply STOP and we won't message about this booking again.

Copy principles: short; the child's name; no fake urgency or guilt; honest about the counsellor call; always an easy way out, because a family that reschedules is recoverable and a family that feels trapped is not.

PART 4 — THE MICRO-EXPERIENCE: BHANZU MATH MISSION

Prototype: `dist/index.html` (single file, opens in any browser) or `npm run dev`. A clearly labelled concept prototype; booking, teacher and slots are sample data and nothing is sent anywhere.

4.1 Why this idea

Covered in 3.2: it is the most valuable uncertainty to resolve, the only child-facing lever, fast to read, and it carries into the demo and the counsellor call. It also delivers ideas 3, 4 and 5 (reschedule, reassurance, joining help) in one place.

4.2 What the parent and child see and do

1. **Home.** A yellow child panel: "Hi Aarav! Your math class is in 3 days." with *Start Math Mission* ("About 5 minutes · Not a test · No sign-up"). Below, the parent card: date, time with time zone (e.g. "5:30 pm IST"), length, teacher, **Confirmed** status, a live countdown, and *How to join*, *Get ready for class*, *Change time*. A timeline: Mission → Live demo → Counsellor call → You decide (no obligation).
2. **Pick your mission.** Three cards explain how it works (5 puzzles, hints if stuck, a badge for the teacher). The child picks an age band (6–8, 9–11, 12–14). No exact age is asked.
3. **Mission.** Five puzzles, easiest first, ending with a speed trick. Progress bar and dots.
4. **Feedback.** Right: ✓ and praise plus a one-line explanation. First miss: ×, "Not quite, and that's okay", a hint, try again. Second miss: the answer and explanation, then move on. Typed answers are validated with child-friendly messages.
5. **If the family is interrupted.** Leaving or refreshing keeps progress; home shows *Continue mission (2 of 5 done)*.
6. **Completion.** Badge, 1–3 stars (never zero), a puzzle-by-puzzle summary, *Play again / Try another level*, and "Tell Ms. Kavya about your badge when class starts on Saturday at 5:30 pm IST."
7. **Parent note.** Which skills needed help, and how the summary would reach the teacher and counsellor in production (not sent in the prototype).
8. **Get ready for class.** What to expect before, during and after; five tips; FAQ including "Will I be pushed to buy something?".
9. **How to join.** Steps, a join button that explains when the real link would open, a readiness checklist, *Add to Google Calendar* (opens Google Calendar prefilled; nothing is saved unless the parent saves it), copyable details, *Can't make it?*.

10. **Change time.** Soonest slots first, a required choice, confirm; the home card shows **Rescheduled** and the new countdown.

Reviewer tools: **Event log** (top right) shows each tracked event with its pseudonymous booking ID; the footer switches the sample booking gap (same day, 1, 3, 5, 7 days); *Reset prototype* clears everything. A link with unreadable booking data (e.g. `?demo=next-tuesday`) shows a notice and a sample booking instead of failing silently.

4.3 The one early signal: 24-hour activation rate

	DEFINITION
Numerator	Treatment bookings in the denominator with at least one `mission_started` event within 24 hours of Message 1 being delivered
Denominator	Eligible treatment bookings whose Message 1 was delivered **and** whose demo starts at least 24 hours after delivery
Window	Fixed 24 hours from delivery, for everyone in the denominator
Short-notice bookings (< 24 h to demo)	Excluded from this metric; reported separately as *pre-demo activation* (started before the demo). Never pooled, because their windows differ
First read	Day 3 of the test: bookings from days 1–2 have complete windows
Activation vs completion	Activation (started) tells us whether delivery and the hook work. Completion (`mission_completed` ÷ starters) tells us whether the content works. Both are tracked; activation is the gate

Why a fixed 24 hours rather than "72 hours or before the demo, whichever is first": that earlier rule gave a 4-day booking three days to activate and a next-day booking a few hours, so the metric would rise or fall with the gap mix instead of with the product. A fixed window makes every booking in the denominator comparable. Message 2 is sent at hour 24, so the 24-hour rate measures the confirmation hook alone.

Decision rules for the day-3 read (thresholds are proposals to agree before launch, not predictions):

- Activation **below 15%**: the hook or delivery is failing. Fix copy, button, timing; restart the cohort clock.
- Activation healthy, **completion below 50% of starters**: the content is too long or hard. Shorten or rebalance.
- Both healthy: hold the test to its planned sample and wait for attendance.

What activation does not prove: families who start the mission are probably more motivated already. A higher attendance rate *among activators* is not evidence of impact. Only the randomized comparison of attendance between arms is.

MEASUREMENT AND EXPERIMENT (SUMMARY)

Full detail: `docs/EXPERIMENT_PLAN.md` and `docs/ANALYTICS_PLAN.md`.

- **Objective:** test whether the Math Mission increases the share of eligible booked demos that are attended.
- **Primary metric:** demo attendance rate, intent-to-treat, using **Bhanzu's existing attendance definition** (the one behind the 65% benchmark), so results are comparable. A stricter "stayed ≥10 minutes" rate is secondary.
- **Arms:** both get the baseline. Treatment adds the mission link, Message 2 and the hub.
- **Randomization:** household (hashed phone number), at booking, 50/50, stratified by gap bucket and market; sticky for the whole test.

- **Reschedules:** a booking keeps its arm and is measured at its final scheduled time, as long as that is within 14 days of the original scheduled time. Cancellations count as not attended.
- **Sample size:** depends on the measured control rate, the minimum lift worth shipping, α, power and weekly volume, none of which Bhanzu has given. Illustration only, at a 65% baseline, α = 0.05, 80% power: +5 points needs about 1,400 bookings per arm; +3 points about 3,900; +2 points about 8,800. The test is not "powered" until real inputs are agreed.
- **Guardrails:** cancellations and reschedules, booking-to-demo flow health, enrollment, opt-outs and complaints, technical failures, counsellor workload, segment differences.
- **Decision matrix:** six pre-agreed outcomes with actions, in `docs/EXPERIMENT_PLAN.md` §10.

PRIVACY AND TRUST

- The child's data in the experience is limited to the first name already in the booking (display only) and a self-selected age band. No exact age, school, photo, voice or free text.
- Analytics use a pseudonymous booking ID; the child's name is never an event property (enforced in code and checked in tests).
- India's DPDP Act requires verifiable parental consent for children's data and prohibits tracking or behavioural monitoring of children. Per-question mission telemetry therefore needs a legal decision before production; the analytics plan proposes keeping only booking-level mission events for the experiment. I have not established compliance and don't claim it.
- Simulated actions are labelled where they happen (join, reschedule, summary sharing) and in a banner on every screen.

RISKS AND LIMITATIONS

- **Selection vs causation:** addressed by randomization and intent-to-treat.
- **Novelty:** read over full weeks; watch activation week over week.
- **Message fatigue:** one extra treatment message; opt-outs are a guardrail.
- **Difficulty:** hints, reveal after two tries, stars never below one, "not a test" framing.
- **Access:** families with one shared phone may not get mission time; the hub still works for the parent.
- **Ops:** rescheduling needs slot supply; "reply HELP" needs staffing at class times.
- **Carry-over:** depends on teachers and counsellors using the summary.
- **Prototype:** sample data; no messaging, booking, class platform, CRM or analytics backend; content not reviewed by Bhanzu teachers or tested with children.

ASSUMPTIONS MOST LIKELY TO BE WRONG

1. That families will open a link in a confirmation message at a useful rate. (Tested by the day-3 read.)
2. That the 65% and the gap table describe the same population and period.
3. That slot supply allows "soonest first" rescheduling.
4. That short-gap bookings dominate volume (see 1.3; needs row counts).

5. That a child's enthusiasm moves a parent's attendance decision.

NEXT STEPS

1. Week 0: breakdowns from 1.5; confirm the attendance definition; legal review of child telemetry.
2. Week 1: ship the baseline.
3. Week 2: launch the mission experiment; daily delivery and error checks.
4. Day 3 of the test: read 24-hour activation against the decision rules.
5. Planned end date: read attendance, secondary metrics and guardrails; decide using the matrix.
6. Afterwards: test the teacher/counsellor summary as its own lever on enrollment; test re-engagement for families who stay silent.